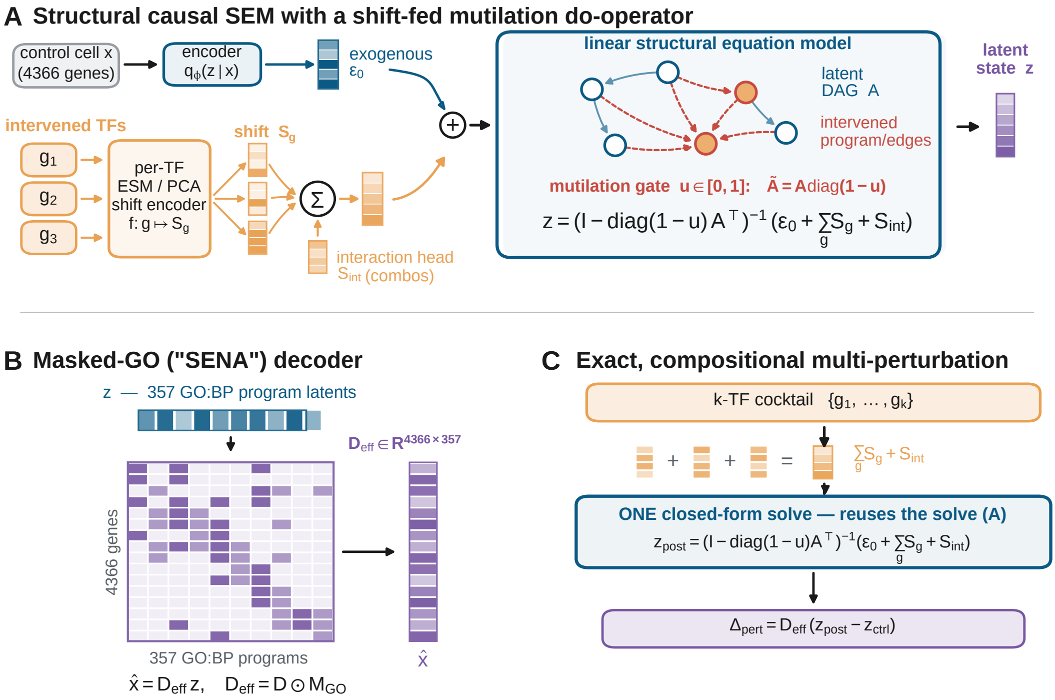
A Structural causal SEM with a shift-fed mutilation do-operator
control cell x (4366 genes)
encoder $q_\phi(z \mid x)$
exogenous ε_0
intervened TFs
g_1
g_2
g_3
per-TF ESM / PCA shift encoder $f: g \mapsto S_g$
shift S_g
Σ
interaction head S_{int} (combos)
$\oplus$
linear structural equation model
latent DAG A
intervened program/edges
mutilation gate $u \in [0,1]$: $\tilde{A} = A\,\mathrm{diag}(1-u)$
$z = (I - \mathrm{diag}(1-u)\,A^\top)^{-1}(\varepsilon_0 + \sum_g S_g + S_{int})$
latent state z
B Masked-GO ("SENA") decoder
z — 357 GO:BP program latents
$D_{eff} \in \mathbf{R}^{4366 \times 357}$
4366 genes
357 GO:BP programs
$\hat{x}$
$\hat{x} = D_{eff}\, z, \quad D_{eff} = D \odot M_{GO}$
C Exact, compositional multi-perturbation
k-TF cocktail $\{g_1, \ldots, g_k\}$
+ + =
$\sum_g S_g + S_{int}$
ONE closed-form solve — reuses the solve (A)
$z_{post} = (I - \mathrm{diag}(1-u)A^\top)^{-1}(\varepsilon_0 + \sum_g S_g + S_{int})$
$\Delta_{pert} = D_{eff}\,(z_{post} - z_{ctrl})$